

Summary: papers 01 and 02

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Two-page summary of two sister manuscripts on data-generating process (DGP) choice and analysis-model specification for biomarker-treatment interactions in aggregated N-of-1 trials under carryover.

Paper 01: Mean moderation and covariance moderation as

data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials, and their divergent power under carryover

Goal. Simulation is the standard tool for power and sample-size calculation in aggregated N-of-1 trials, but any such simulation must specify how the biomarker moderates the treatment effect in its DGP, and this specification is a substantive scientific choice rather than an implementation detail. Each of the two established parameterizations carries its own advantages and costs. The paper formalizes both and quantifies how much the choice between them changes the estimated power loss under carryover.

Under *mean moderation*, the biomarker scales the treatment effect in the conditional mean, a deterministic relationship appropriate when the biomarker directly determines drug pharmacokinetics or pharmacodynamics. Under *covariance moderation*, the interaction is represented as a correlation between biomarker and biological response that depends on drug-exposure condition (on drug versus off drug), a probabilistic relationship appropriate when the biomarker is only a proxy for a latent responder subtype. Covariance moderation is often the more biologically defensible choice in that latter setting, and it is the specification Hendrickson et al. selected for the motivating prazosin and PTSD application on exactly these grounds.

Method. Both architectures are embedded in an otherwise identical Monte Carlo simulation (linear mixed model with continuous-time AR(1) residual correlation) and compared across three within-subject designs (crossover, Hybrid, and open-label with blinded discontinuation) matched to the prazosin and PTSD reference application, as carryover half-life increases from zero to one week.

Results. Under mean moderation, power is nearly preserved because carryover does not disrupt the biomarker's relationship to the treatment effect: in the OL+BDC design at $N = 70$, power falls only from 0.751 to 0.730 (a 2.8% relative loss). Under covariance moderation, the same carryover erodes the differential correlation that carries the signal, and power falls from 0.777 to 0.539 (a 30.6% relative loss), roughly eleven times the mean-moderation loss and far beyond Monte Carlo error ($z = 7.64$, $p < 10^{-13}$). The gap is strongly design-dependent: large in designs with a discontinuation phase, and negligible in a classical crossover, because the within-subject AR(1) correlation there already carries most of the signal that covariance moderation depends on, leaving carryover little left to erode.

Conclusion. The DGP architecture is a substantive scientific assumption, not a parameterization convenience, and it determines how severely carryover appears to reduce power. That defensibility does not eliminate a real carryover cost when covariance moderation is the biologically appropriate choice. Biomarker-validation simulation studies should state which architecture they adopt and report power under both.

Paper 02: Unadjusted, lag-adjusted, and exposure-weighted

carryover specifications under differential-correlation biomarker interactions

Goal. Paper 01 establishes that carryover costs up to roughly 31% of power under covariance moderation. Given that architecture, the analyst must still decide how to represent the resulting residual drug exposure in the analysis model, and it is not established which representation is most robust when the true pharmacokinetic decay shape is unknown, as is ordinarily the case in practice. Paper 02 takes up that question directly.

Three analysis-model specifications for the drug-exposure regressor are compared. *Unadjusted* is the binary on-drug indicator, ignoring carryover entirely and serving as the benchmark. *Lag-adjusted* augments that indicator with a lagged just-off-drug term that estimates carryover magnitude non-parametrically, the classical crossover-trial device. *Exposure-weighted* is a continuous indicator that commits to a parametric decay form and half-life, the current default in the pmsimstats-ng package.

Method. All three specifications are fitted to the same simulated datasets under an ADEMP-structured factorial crossing two DGP decay forms (exponential, Weibull) with the three specifications, three trial designs, two sample

sizes, and three interaction strengths (216 cells, 500 replicates each), together with four single-axis sensitivity analyses and a cluster-robust recalibration.

Results. Two findings stand out. First, the lagged-treatment term buys nothing: Lag-adjusted and Unadjusted are statistically indistinguishable across all 216 cells, agreeing to three decimal places on bias, mean squared error, and coverage. Because the two share an estimand and differ by exactly one nuisance term, this is a clean null, traced to the lagged term entering the model only as a nuisance main effect that leaves the attenuation of the interaction untouched. Second, Exposure-weighted attains the highest power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage is about ten percentage points (0.860 versus 0.766 for Unadjusted); under the classical crossover design it is markedly inferior (0.488 versus 0.830), because the within-subject correlation there already carries most of the signal the decaying predictor was designed to recover. Where Exposure-weighted leads, the lead is robust to decay-shape and half-life mis-specification and to autocorrelation strength. The model-based interaction test is mildly conservative for all three specifications, an artifact of the working covariance that a cluster-robust standard error removes without disturbing the ranking. Across all conditions, anticipated dropout reduces power far more than the choice among specifications.

Conclusion. Under covariance moderation, we recommend the exposure-weighted predictor, reported with a cluster-robust standard error, for the Hybrid and OL+BDC designs; under a crossover design it should not be used, and the unadjusted binary indicator is materially better. The classical lagged-treatment remedy cannot be recommended in any design examined. Preventing dropout is a more effective use of design effort than any choice among these specifications.

How the two papers relate

Paper 01 establishes that the choice of DGP architecture, not just the analysis model, determines how severely carryover erodes power, and that covariance moderation, often the more biologically defensible choice, pays the larger cost. Paper 02 holds that architecture fixed and asks the practical follow-up question: given that a trial is analyzed under covariance moderation, which representation of residual drug exposure in the analysis model best recovers the power carryover would otherwise cost. Together, the two papers argue that both the data-generating process and the analysis-model specification are substantive design choices that should be reported and justified, not left to implementation convenience.